

20	A	Let current in the circuit be I. $I = \frac{Q}{t} = \frac{Ne}{t}$ N, number of electrons = $(8.5 \times 10^{28}) \times \text{volume of wire}$ $= (8.5 \times 10^{28}) \times (3.2 \times 10^{-7})(0.0047)$ $I = \frac{(8.5 \times 10^{28})(3.2 \times 10^{-7})(0.0047)(1.60 \times 10^{-19})}{(60)}$ $= 0.34$
21	D	$3R$